

OSIPI · GSOC 2026 · ASL QUALITY CONTROL

Kidney QC ToolBox Renal ASL QC Design

Automatic PASS / WARN / FAIL triage for **renal** ASL MRI

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KIDNEY_QC_DESIGN.md · 14,593 words

reconciled against osipy-qc · 20 registered checks · 253 tests

Overview

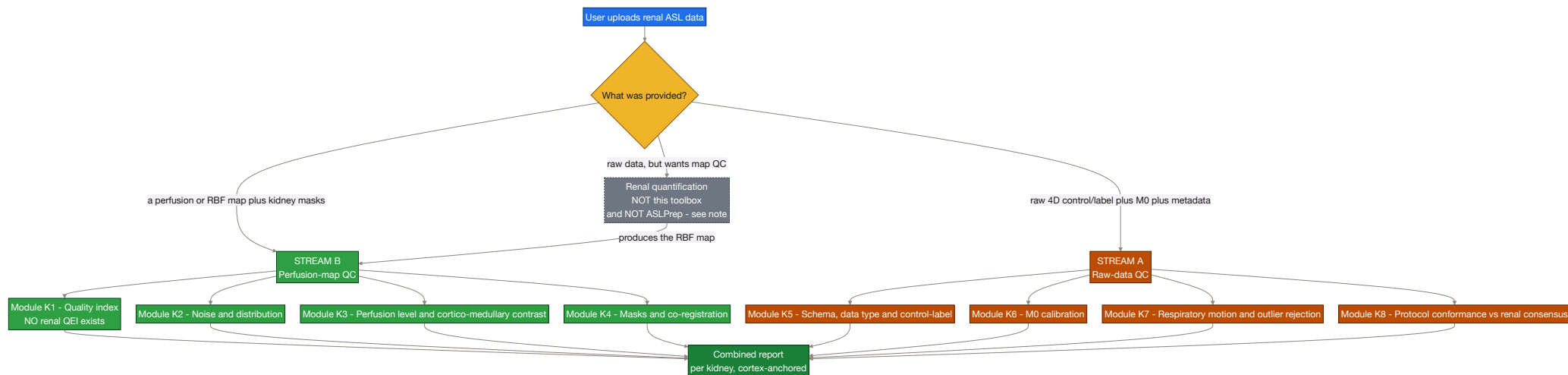
This is the renal counterpart to `QC_DESIGN.md`. It keeps the same shape — two streams, eight modules, one spec card per check — so the brain and kidney toolboxes can share a registry, a result type, and a report format. What it does **not** keep is the brain's evidence base.

The single most important fact about this design: the renal ASL literature contains a rich, quotable, consensus-graded specification for **how a scan should be acquired**, and almost nothing at all about **how to judge whether the resulting map is good**. The one authoritative document — Nery et al. 2020 ([nery2020_renal_asl_consensus.pdf](#), MAGMA 33:141–161, PARENCHIMA COST CA16103, 23-expert modified Delphi, 59 agreed statements) — contains **zero numeric image-quality thresholds**: no tSNR floor, no CoV cutoff, no motion limit, no negative-voxel fraction, no QEI analogue. Its only two QC-adjacent statements are R8.1 ("retrospective image registration is highly recommended", 100% agreement) and R8.2 ("outlier rejection is recommended", 100% agreement), and **both specify a method of exactly zero and a threshold of exactly zero**.

That produces an inversion relative to the brain toolbox, and it should be stated plainly to the mentors rather than hidden:

	BRAIN V1.0	KIDNEY
Stream A (was the acquisition correct?)	thin — White Paper Table 1, a handful of rules	rich — ~25 consensus statements with agreement percentages
Stream B (is the perfusion map good?)	rich — QEI with a validated ≈ 0.5 cutoff, sCoV 0.67, GM/WM 40–100	almost empty — no renal QEI, no renal sCoV, no published band

So the honest v1 kidney toolbox is **Stream-A-heavy**. Stream B ships real, computable metrics, but almost every renal Stream-B threshold is `UNCALIBRATED`, and this document says so at every single one rather than manufacturing bands to look symmetrical with the brain doc.



⚠️ **The gray dashed box is emptier than it is for brain.** For brain, "the pipeline makes the map" means ASLPrep / ExploreASL / PyASL. For kidney **there is no such pipeline**. I grepped the local clones: ASLPrep, ExploreASL and PyASL contain **zero** kidney/renal code (ASLPrep's only **renal** string match is a false positive inside a base64 blob in `docs/_static/brainextraction.svg`). The renal MRI community's flagship toolbox, UKRIN's `ukat`, has **no ASL module at all** — its mapping subpackage is B0/diffusion/MTR/T1/T2/T2*, and its entire QA subpackage is one `snr.py`. There is an unreleased 4.8 kB `perfusion.py` on `ukat`'s `dev` branch that does label-control subtraction with no M0, no quantification and no QC; it opens with `warnings.warn('Perfusion is still a work in progress and likely to be re-written. Use with caution.')`. **Consequence for scope:** the kidney toolbox cannot assume it will be handed a pipeline's derivatives. It must consume near-raw NIFTI plus user-supplied masks.

Positioning inside OSIPi. OSIPi has no renal task force, no renal lexicon section beyond three kidney rows in the living Google-Doc lexicon, no renal reporting checklist and no renal ASL datasets (TF3.2's inventory has one kidney entry and it is DCE, and the task force page says it "is currently inactive due to insufficient members"). The authoritative renal ASL standard is PARENCHIMA's, not OSIPi's. But both OSIPi task forces whose roadmap explicitly targets non-brain ASL are led by this project's own mentors — **TF2.2 (ASL code library) by María Mora Álvarez**, **TF1.1 (pipeline inventory) by Sudipto Dolui** — and the OSIPi roadmap states in writing: "We do not have any pipelines able to process human clinical non-brain data ... there is no large free available pipeline for processing these data." This work fills a gap OSIPi has already documented and owns.

⚖️ How the verdict works

Identical machinery to brain — same `Verdict` enum, same aggregation — with **two renal-specific additions**:

- Everything is reported per kidney.** Consensus R10.1 (0% abstentions, **100% agreement**): "Cortical renal blood flow values (not whole-kidney) should be reported, separately for left and right kidney." So the report object is `{left: {...}, right: {...}}`, not a single scalar, and the aggregator runs per kidney then combines. This is the strongest, cleanest published constraint in the whole renal corpus and it should drive the data model.
- Cortex is the anchor ROI; medulla is INFO-only.** Consensus R10.2 (14% abstentions, 89% agreement): "Medullary renal blood flow values are not considered reliable with current measurement approaches." Backed by measurement: medullary inter-visit ICC spans 0.13–0.96 with CV 4–37% ([odudu2018_renal_asl_systematic_review.pdf](#)), and Garcia-Ruiz measured a medullary inter-session ICC of **0.08 at 3T** — essentially no subject-specific information at all ([garciaRuiz2025_field_strength_sex_age_kidney.pdf](#)). **No medullary metric may drive a FAIL.**

Aggregation is unchanged: any FAIL → FAIL; else any WARN or UNKNOWN → WARN; else PASS. **N/A** and **INFO** are excluded. One added rule, inherited from `THRESHOLD_PROVENANCE.md` and now doing much more work: **an uncalibrated threshold never drives a FAIL on its own**. In the brain toolbox that rule protected a minority of checks. Here it protects nearly all of Stream B.

Provenance tiers used throughout

TAG	MEANING	HOW COMMON IN THIS DOC
PUBLISHED	a peer-reviewed paper states this number, for this purpose	mostly Stream A protocol conformance
IMPLEMENTATION	a reference study/pipeline's code or method uses it; not a validated cutoff	the outlier-rejection rules, the 0–500 clip
UNCALIBRATED	engineering default, no published source	most of Stream B
DEFINITION	physics/math, nothing to tune	ratios, centroids, formulas

Consensus statements are quoted **with their agreement and abstention percentages**, because they vary a lot and a 76% statement should not be enforced as hard as a 100% one. Highest-abstention warning: R5.10 (Gmax/Gave) carries **40% abstention** and the paper contradicts itself between Table 2 ("6–7") and the Discussion ("of 7") — treat as advisory only.

📁 Minimum inputs — what you must bring for each tier of kidney QC

The mentors asked for this explicitly for the brain tool; it will be asked again, and the renal answer is different enough to be worth its own table. **Nothing here can be auto-derived the way a brain mask can** — consensus R9.10 (0% abstentions, 100% agreement) makes **manual ROI the default**, and R9.11 (93%) says ROIs should be drawn on the **ASL M0 image or a separately acquired structural dataset**, not on the perfusion map.

TIER	YOU MUST SUPPLY	WHAT BECOMES COMPUTABLE	WHAT STAYS UNKNOWN
B0	perfusion / RBF map only	almost nothing — global histogram shape, implausible-value fraction	every ROI metric; there is no renal analogue of "just BET the brain"
B1	+ whole-kidney masks, left and right separately	negative-voxel fraction, implausible-value fraction, whole-kidney level (reported but <i>not</i> the consensus quantity), left-vs-right consistency, mask-integrity checks	cortical RBF (the consensus-mandated value), cortex:medulla ratio
B2	+ cortex mask per kidney	cortical RBF level per kidney — the quantity R10.1 requires; cortex-restricted CoV; cortical negative fraction	cortex:medulla ratio
B3	+ medulla mask per kidney (or a quantitative T1 map to derive one)	cortex:medulla ratio as a segmentation-integrity flag (INFO/WARN only, never FAIL)	—

TIER	YOU MUST SUPPLY	WHAT BECOMES COMPUTABLE	WHAT STAYS UNKNOWN
A0	raw 4D control/label series + kidney masks	subtraction-outlier rate, per-kidney respiratory displacement, control/label ordering, surviving-pair count	everything metadata-driven
A1	+ M0 / PD image	M0 presence (mandatory per R9.1), ASL↔M0 registration, PWS as % of M0, background-suppression efficiency	M0 TR rule
A2	+ acquisition metadata (JSON sidecar or vendor header)	the whole of Module K8 — the richest published block in this design — plus M0 TR ≥ 5 s	—
A3	+ respiratory trace (<code>_physio.tsv.gz</code>)	gating efficiency, motion-vs-outlier correlation	—

Practical reading of this table: the highest-value single input a renal user can add is **cortex masks per kidney** (B1 → B2), because it unlocks the one quantity the consensus actually asks you to report. The highest-value *metadata* input is a JSON sidecar, because Module K8 is where nearly all the published thresholds live. And a quantitative **T1 map** is disproportionately valuable: it is the only contrast that reliably separates cortex from medulla (it is what `bones2022_automatic_renal_perfusion_workflow.pdf` 's third U-net segments on), and it supplies the `T1,tissue` that the consensus M0 correction equation needs but never numerically specifies.

⚠ **Auto-segmentation is possible but must be advisory.** Renal Segmentor (the engine behind `ukat`) reports whole-kidney Dice 0.93 ± 0.01 and can emit a per-voxel probability map — the closest renal analogue to a brain tissue prior. Two caveats: it is **whole-kidney only**, and that 0.93 validates the *original 2021* network, while its README says the shipped weights have since been **retrained on 378 CKD subjects across three vendors**, so 0.93 is a superseded number. For cortex/medulla there is no shipped tool at all: `ukat` promised automated cortico-medullary segmentation "by the end of 2021" in its ISMRM abstract and current `master` still contains **zero** matches for `cortex` or `medulla` in code.

● STREAM B — QC of the perfusion map (*Modules K1–K4*)

🔪 Shared PRE-STEP

1. **Split by kidney.** Every downstream statistic is computed on left and right independently. Never pool. (R10.1, 100%.)
2. **Resample masks to the perfusion grid.** Same as brain, but note the renal grid is coarse and anisotropic by design: consensus recommends in-plane 2–4 mm (R6.10, 93%) with 4–8 mm slices in 2D (R6.8, 100%).
3. **No default tissue priors.** The brain pre-step falls back to warped MNI priors when no segmentation is supplied. **There is no renal equivalent** — no standard renal atlas space, no cortex/medulla priors, and kidney position/shape varies far more than brain. If masks are absent, ROI checks return UNKNOWN. This is a real capability loss versus the brain toolbox and should be said out loud.
4. **No smoothing step.** Brain pre-step smooths at 5 mm FWHM *inside* the QEI engine because Dolui's constants were fitted on 5 mm-smoothed data. **There is no renal QEI, therefore no renal smoothing prescription.** Smoothing a 2–4 mm in-plane renal map with a 5 mm kernel would also blur straight across a cortex that can be only a few voxels thick. Do not port this step.
5. **Clean non-finite voxels** — unchanged from brain.
6. **Record the acquisition context on every metric.** Every Stream-B number is emitted with `{labelling_scheme, field_strength_T, pld_or_ti_s, readout, subject_age, native_or_transplant}` attached, because — see K3.1 — those factors move renal cortical RBF more than disease does.

● Module K1 — Quality index ★ (*the QEI slot — and why it is empty*)

K1.1 Renal composite quality index • NOT IMPLEMENTED IN v1

🔗 **what it would check:** a single 0–1 number summarising renal perfusion-map quality, the way QEI does for brain.

📌 **verdict:** always `N/A` in v1, with an explanatory reason string. It is deliberately present in the design as a named, empty slot rather than silently omitted, so that reviewers can see the gap was identified rather than overlooked.

Why it cannot be ported — four independent blockers:

1. **No tissue-probability substrate.** QEI's structural term is $\text{spCBF} = 2.5 \cdot \text{GM_prob} + 1.0 \cdot \text{WM_prob}$ and $\text{p_ss} = \text{Pearson}(\text{CBF}, \text{spCBF})$. The renal analogue would be $\text{spRBF} = a \cdot \text{cortex_prob} + b \cdot \text{medulla_prob}$. There is no consensus `a:b` — measured cortex:medulla ratios cluster at 2.26–2.65 in cleanly-segmented studies, while the XCAT synthetic phantom (`brumer2022_synthetic_renal_asl_phantom.pdf`) assumes 5.0 as an input, citing Roberts 1995. An assumption cannot validate itself.
2. **The boundary is not reliably drawable, even by humans.** Automatic cortex segmentation reached Dice 0.78 ± 0.04 against reference — and a second human expert reached 0.77 ± 0.02 against the same reference (`bones2022_automatic_renal_perfusion_workflow.pdf`). Human inter-observer agreement on the renal cortex boundary is ~ 0.77 , against >0.90 typical for brain GM/WM. Note the inference limit: a 0.78 Dice produced only a **1.4%** difference in cortical RBF in that same paper, so Dice deficit does **not** convert into a proportional metric error — but it does mean the mask is not a stable enough scaffold for a fitted structural-correlation term.

3. **The denominator compartment is declared unreliable** by the consensus itself (R10.2, 89%), so any pooled-variance term across cortex and medulla — QEI's **DI** analogue — inherits that.

4. **No labels, therefore no curves and no cutoff.** QEI's f_1 f_2 f_3 curves and its ≈ 0.5 cutoff were fitted against expert ratings on ~ 500 scans. **I found no renal ASL dataset with expert quality ratings.** Without labels there is nothing to fit and nothing to threshold. A QEI-Net analogue is even further out for the same reason.

Research direction (stretch, needs mentor input — see open question 6): the pieces of a renal quality index that are individually computable today are shipped as separate checks in K2–K4 (structural plausibility via the cortico-medullary ratio, dispersion via cortex-restricted CoV, negativity via the negative-voxel fraction). Combining them into a composite is premature: a geometric mean of three uncalibrated components is not more trustworthy than its components, it is less interpretable. **Ship the components; defer the composite until a labelled renal dataset exists.**

provenance: the *absence* is the finding. verified by exhaustive term scan of [nery2020_renal_asl_consensus.pdf](#) (zero occurrences of **quality control**, **quality assurance**, **CoV**, **cut-off**, **criteri***, **DVARS**, **framework**; **tSNR** appears once and only qualitatively) and by Europe PMC searches returning no renal quality index.

Module K2 — Noise & distribution

K2.1 Temporal SNR (tSNR) • REQUIRED (report-only in v1)

what it checks: how stable the perfusion-weighted signal is across repetitions.

inputs:

```
{
  "delta_m_4d": "NIfTI 4D float", # per-repetition control-label subtractions
  "kidney_masks": {"left": "NIfTI 3D bool", "right": "NIfTI 3D bool"},
  "cortex_masks": {"left": "NIfTI 3D bool", "right": "NIfTI 3D bool"}, # optional, preferred
  "n_pairs": 25, # number of control/label pairs averaged
}
```

output:

```
{
  "metric": {
    "left": {"tsnr_cortex": 3.4, "tsnr_whole": 2.1, "n_pairs_used": 24},
    "right": {"tsnr_cortex": 3.1, "tsnr_whole": 2.0, "n_pairs_used": 23},
    "definition": "mean_dM_over_time / sd_dM_over_time, per voxel, averaged in ROI",
    "per_repetition": True, # NOT the SNR of the averaged map
  },
  "verdict": "INFO | UNKNOWN",
  "reason": "cortical tSNR 3.4 (L) / 3.1 (R) – reported, no calibrated threshold exists",
}
```

how I plan to compute it (method):

- Voxelwise `mean( $\Delta M$ , axis=t) / std( $\Delta M$ , axis=t)` inside each ROI, averaged over ROI voxels — the definition used in [hartevelde2020_multidelay_fair_vs_pcasl.pdf](#) and [garciarui2025_field_strength_sex_age_kidney.pdf](#).
- Emit the definition string and `per_repetition: True` in the metric itself, because renal tSNR is not comparable across papers otherwise (see the warning below).
- Report per kidney, cortex-restricted where a cortex mask exists.

PASS / WARN / FAIL: none in v1. This check emits **INFO**. A threshold would be irresponsible — see below.

thresholds & sources:

- tSNR definition** — definition; stated verbatim in [hartevelde2020_multidelay_fair_vs_pcasl.pdf](#) and [garciarui2025_field_strength_sex_age_kidney.pdf](#).
- Reference values (context only, NOT thresholds):** cortex 3.54 ± 0.71 at 1.5 T and 3.33 ± 0.54 at 3 T (n=16, PCASL, SE-EPI, background-suppressed — [garciarui2025](#)); cortex 1.5–2.6 depending on readout (n=10, 3 T FAIR — [buchanan2018_2d_imaging_schemes_kidney_cortex.pdf](#)); whole kidney 0.60 ± 0.15 without background suppression rising to 0.93 ± 0.22 with heavy BS (n=9, 1.5 T pCASL — [bones2019_freebreathing_bgs_renal_pcasl.pdf](#)); medulla 0.17 ± 0.14 for VSI-ASL ([franklin2021_multiorgan_flowbased_asl.pdf](#)).
- PASS/WARN/FAIL band** — **UNCALIBRATED** and deliberately not set.

Two traps that would each produce a badly wrong renal tSNR threshold.

(1) **The definitions are not the same quantity.** Buchanan defines tSNR as mean PW signal divided by the SD across the 25 ASL pairs — a **per-repetition** figure. The averaged perfusion map built from 25 pairs has roughly $\sqrt{25} = 5\times$ higher SNR. A threshold copied from "tSNR 1.5–2.6" onto an averaged map would be about $5\times$ too strict and would fail nearly every good kidney scan. (2) **The values span $\sim 7\times$ across published studies** ($0.60 \rightarrow 4.61$) driven by ROI, field strength, labelling scheme, readout and repetition count — not by data quality. tSNR is only comparable within a fixed (definition, ROI, readout, labelling, field-strength, n_pairs) tuple, which is why the metric carries all of those as attached context.

🔗 **needs:** a 4D ΔM series and at least a whole-kidney mask. Without a series → UNKNOWN (a single averaged map has no temporal axis).

🔗 **catches:** unstable subtraction — the joint signature of respiratory motion, labelling failure and inadequate background suppression.

K2.2 Perfusion-weighted signal as % of M0 • REQUIRED

🔗 **what it checks:** is the ASL difference signal of a physically sensible magnitude relative to the calibration image? This is the renal analogue of asking "does this look like ~1% ASL contrast?"

🔗 **inputs:** `delta_m` (mean subtraction), `m0`, `kidney_masks`, `cortex_masks` (optional), `n_bs_pulses` (optional).

🔗 **output:**

```
{
  "metric": {
    "left": {"pws_cortex_pct": 3.0, "pws_medulla_pct": 1.4},
    "right": {"pws_cortex_pct": 2.9, "pws_medulla_pct": 1.3},
    "bs_efficiency_pct": None,      # only if a no-BS reference acquisition exists
  },
  "verdict": "PASS | WARN | UNKNOWN",
  "reason": "cortical PWS 3.0% – within the reported renal range",
}
```

🔗 **method:** $PWS\% = 100 \cdot \text{mean}(\Delta M[\text{roi}]) / \text{mean}(M0[\text{roi}])$, per kidney, cortex-restricted where possible. Requires ΔM and M0 to be in the same space (gate on check K4.2).

📊 **PASS / WARN / FAIL:** | outcome | condition | |---| | PASS | cortical PWS in ~0.5–8% | | WARN | outside that, or ≤ 0 (points at a labelling or subtraction-sign failure) | | UNKNOWN | no M0, or ΔM and M0 not co-registered |

📌 **thresholds & sources:**

- **Reference values:** cortex $2.95 \pm 0.56\%$ (1.5 T) and $3.09 \pm 0.59\%$ (3 T); medulla $1.36 \pm 0.15\%$ and $1.40 \pm 0.19\%$ — [garciarui2025_field_strength_sex_age_kidney.pdf](#), n=16. Whole kidney $3.43 \pm 1.43\%$ in paediatric allografts — [radovic2022_paediatric_allograft_asl.pdf](#). Cortex peak $5.98 \pm 0.70\%$ (FAIR at optimal TI) — [bones2021_vsasl_label_dynamics_kidney.pdf](#). Paediatric contralateral kidney $1.5 \pm 0.61\%$ at PLD 0.5 s falling to $0.51 \pm 0.27\%$ at PLD 1.5 s — [hartevel2022_paediatric_neuroblastoma_nephroblastoma_pcasl.pdf](#). Review-level statement "typical renal cortex perfusion-weighted signal intensity of 5% of the control image signal" — [odudu2018_renal_asl_systematic_review.pdf](#).
- **The 0.5–8% band** — 🔗 **UNCALIBRATED.** It is drawn to contain the full measured spread above, not fitted to anything. Note the measured cortical values (~3%) sit *below* the review's oft-quoted 5%, so a band anchored on 5% would be wrong. **PWS is strongly PLD-dependent** (1.5% → 0.51% across PLD 0.5–1.5 s in the same children), so the check must be PLD-conditioned before it could ever become a FAIL.

🔗 **catches:** labelling failure, wrong subtraction sign, M0 scaling errors, and background suppression that is far more or less aggressive than declared.

K2.3 Implausible-value and negative-voxel fraction • REQUIRED

🔗 **what it checks:** what fraction of within-kidney voxels are physiologically impossible.

🔗 **inputs:** `rbf_map`, `kidney_masks`, `cortex_masks` (optional), `units`.

🔗 **output:**

```
{
  "metric": {
    "left": {"neg_frac_cortex": 0.03, "over_500_frac": 0.004, "n_voxels": 812},
    "right": {"neg_frac_cortex": 0.05, "over_500_frac": 0.011, "n_voxels": 790},
    "clip_units": "mL/100g/min",
  },
  "verdict": "PASS | WARN | FAIL | UNKNOWN",
  "reason": "3–5% negative cortical voxels",
}
```

🔗 **method:**

1. Negative fraction = $\#(\text{rbf}[\text{cortex}] < 0) / \# \text{cortex}$.
2. Implausible-high fraction = $\#(\text{rbf}[\text{cortex}] > 500) / \# \text{cortex}$.
3. **Unit guard before anything else.** The published clip is stated per **100 g** at origin (Perfusion values outside 0–500 mL (100 g)⁻¹ min⁻¹ were rejected, Gullaksen 2023, Diabetologia) and is re-cited per **100 mL** by [olsen2025_renal_asl_vs_o15_pet.pdf](#). These are different normalisations. Refuse to apply the clip unless units are declared; emit UNKNOWN otherwise.

📊 **PASS / WARN / FAIL:** | outcome | condition | |---| | PASS | negative cortical fraction < ~5% and implausible-high fraction < ~5% | | WARN | 5–20% on either | | FAIL | > 20% negative — points at a control/label swap or a failed subtraction | | UNKNOWN | no mask, or units not declared |

📌 **thresholds & sources:**

- **The 0–500 mL/100 g/min voxel window** — **IMPLEMENTATION.** Real and published (Gullaksen 2023, applied in [olsen2025_renal_asl_vs_o15_pet.pdf](#)), but it is a per-voxel **preprocessing clip** inside one study's segmentation step, not a validated scan-level quality bound. The rodent protocol chapter's display range `[0 500]` is explicitly prefaced "for example" and is **not** independent corroboration. **Implement it two-sided (0 and 500), because the original rule rejects negatives too** — that half is the direct renal analogue of the brain's negative-voxel check and is more useful than the upper clip.
- **5% / 20% fraction cut-points** — **UNCALIBRATED.** Nobody publishes "reject the scan if X% of cortical voxels are negative." Carried over in spirit from the brain module, not from renal evidence.
- **Physiological ceiling context:** whole-kidney flow is ~1200 mL/min ≈ **400 mL/100 g/min** in a 70 kg adult with a 300 g kidney ([odudu2018](#)), so cortical voxels far above ~500 imply more than the entire renal blood supply. Note this is a *whole-kidney mass-normalised* physiology figure — do not compare it directly to cortical ASL means.

catches: control/label swap, subtraction sign errors, vascular contamination (bright focal outliers), and gross quantification-constant errors.

K2.4 Cortex-restricted spatial CoV • **STRETCH (exploratory – no renal precedent)**

what it would check: spatial heterogeneity of cortical perfusion, as a possible renal analogue of the brain sCoV / transit-artefact indicator.

verdict: **INFO** only. Never PASS/WARN/FAIL in v1.

provenance — this is a genuine negative result and should be reported as one:

- **UNCALIBRATED, and worse: no renal precedent exists.** Europe PMC searches for ("spatial coefficient of variation" OR "spatial CoV") AND (renal OR kidney) returned 21 hits and "spatial coefficient of variation" AND "arterial spin" returned 88 — **every relevant hit is cerebral.** The kidney hits are brain perfusion studies in kidney-transplant recipients. The brain sCoV → transit-artefact logic has never been published for kidney.
- **Why a naive whole-kidney sCoV would be actively misleading:** the kidney has an intrinsic cortico-medullary perfusion gradient of ~2.3–2.6× (K3.4). A whole-kidney CoV would be dominated by **normal anatomy**, not artefact. If this metric is computed at all it must be **cortex-restricted**.
- **Do not calibrate this on Gillis's 9.2%.** [gillis2014_interstudy_reproducibility_3t.pdf](#) reports a within-subject **between-session test-retest** CoV of 9.2% for cortical perfusion (ICC 0.85). That is a repeatability statistic across two scan days, **not** a spatial CoV across voxels within one map. They share a name and nothing else, and they differ by roughly an order of magnitude. A renal spatial-CoV threshold set at 9.2% would fail essentially every real kidney map.
- **Possibly a better renal primitive:** the SAMURAI 5 T-vs-3 T study reports a **cortico-medullary contrast-to-noise ratio** (CNR 3.2 at 5 T vs 2.0 at 3 T, n=20). CNR between compartments may be a more natural renal heterogeneity metric than CoV within one. Worth asking the renal specialist — see open question 9.

Module K3 — Perfusion level & cortico-medullary contrast

K3.1 Cortical RBF level, per kidney • **REQUIRED**

what it checks: is cortical renal blood flow in a physiologically plausible range — the consensus-mandated reported quantity (R10.1, 100%).

inputs: `rbf_map`, `cortex_masks` (per kidney), plus the acquisition context `{labelling_scheme, field_strength_T, subject_age, native_or_transplant, units}`.

output:

```
{
  "metric": {
    "left": {"cortical_rbf_mean": 312.0, "cortical_rbf_sd": 48.0, "cortical_rbf_median": 305.0},
    "right": {"cortical_rbf_mean": 298.0, "cortical_rbf_sd": 51.0, "cortical_rbf_median": 291.0},
    "band_used": "none – reported against context, not thresholded",
    "context": {"labelling": "FAIR", "field_T": 3.0, "age": 51, "graft": False},
  },
  "verdict": "INFO | WARN | UNKNOWN",
  "reason": "cortical RBF 312 / 298 mL/100g/min; outside any published per-scan reference interval",
}
```

method: mean, SD and median of the RBF map inside each cortex mask. Median is included because the consensus explicitly asks for it: "*The median should also be considered in the presence of skewed RBF distributions.*" Report left and right separately, never pooled.

PASS / WARN / FAIL: v1 emits **INFO** plus a wide implausibility **WARN** only.

OUTCOME	CONDITION
INFO	value reported with its acquisition context (the default)
WARN	cortical mean outside ~50–500 mL/100 g/min — a sanity bound, not a normal range
UNKNOWN	no cortex mask, or units undeclared

thresholds & sources — read this section before proposing any tighter band:

- **Published literature spread:** 139–427 mL/100 g/min in healthy volunteers, 83–412 in patients — [odudu2018_renal_asl_systematic_review.pdf](#), verbatim. **This is a range of study-level cohort MEANS across 53 heterogeneous studies** (mean 25 ± 23 participants each, mixed 1.5/3 T, mixed

FAIR/pCASL, mixed quantification constants). It is **not** a per-scan reference interval, and the healthy and patient ranges overlap almost entirely — so it has near-zero discriminative power as a PASS band. Using it as one would be the cohort-mean-as-threshold error.

- **Individual-subject range, PET-validated:** 150–422 mL/min/**100 mL** for ASL vs 184–470 for [¹⁵O]H₂O PET, n=10 healthy, 3 T FAIR — [olsen2025_renal_asl_vs_o15_pet.pdf](#). This is an individual-level range and is the better anchor, but note the units are **per 100 mL (volume)**, not per 100 g. Agreement with PET was acceptable only between ~250 and ~350; overall bias 18 with limits of agreement ±136 mL/min/100 mL. **The reference standard itself disagrees by ±136** — the strongest single argument that no accuracy-based band is defensible.
- **The ~50–500 sanity bound** — **UNCALIBRATED**. Deliberately wide.

Technical factors move this number more than disease does. Any band must be keyed on acquisition before population.

- **Labelling scheme: ~1.8x.** In the *same 16 subjects, same session*, FAIR gave cortical RBF 362 ± 57 vs pCASL 201 ± 72 mL/min/100 g (P < 0.001) — [hartevel2020_multidelay_fair_vs_pcasl.pdf](#). This is larger than any healthy-vs-diseased, native-vs-transplant or adult-vs-child difference in this entire evidence base.
- **Field strength: ~11%.** Same subjects, same day: cortex 347.65 ± 54.08 at 1.5 T vs 310.63 ± 52.72 at 3 T (P < 0.001) — [garciarui2025](#). Note the counter-intuitive direction (higher field, *lower* measured RBF).
- **Age: ~20%.** 1.5 T single-delay cortex 383.90 ± 61.80 for <40 y vs 306.10 ± 41.33 for ≥40 y — [garciarui2025](#).
- **ROI convention: ~7%.** Two competent human observers differed by 14 mL/min/100 g ≈ 7% on the same data — [bones2022](#).
- **Quantification constants:** [cox2017_multiparametric_renal_mri_validation.pdf](#) uses λ = 0.8 and T1b = 1.55 s at 3 T; the consensus specifies λ = 0.9 and 1.65 s. That alone is a ~12% offset before any physiology.

Reproducibility bounds how tight *any* renal band can be: cortical intra-visit ICC 0.62–0.98 with CV 3–18%, inter-visit ICC 0.85–0.97 with CV 4–13% ([odudu2018](#), from the 17 of 53 studies reporting it).

catches: gross quantification errors, wrong constants, wrong units, mask placed outside the kidney.

K3.2 Cortico-medullary ratio (CMR) • **REQUIRED** (as a segmentation-integrity flag, not a perfusion verdict)

This is the check the mentors will most want to discuss, so it is specified carefully — including an explicit statement of what it is **not** allowed to conclude.

what it checks: $CMR = \text{mean}(RBF[\text{cortex}]) / \text{mean}(RBF[\text{medulla}])$, per kidney. It is proposed as the renal analogue of the brain GM/WM ratio in *form* — scale-free, robust to global scaling errors, independent of quantification constants — but **not in role**. In brain, GM/WM ratio < 1 is a published data-quality FAIL (Adebimpe 2022). In kidney, CMR is proposed as a **flag on the masks, not on the kidney**.

inputs: `rbf_map`, `cortex_masks`, `medulla_masks` (per kidney), `t1_map` (optional, to derive masks).

output:

```
{
  "metric": {
    "left": {"cmr": 2.41, "cortex_mean": 312.0, "medulla_mean": 129.5},
    "right": {"cmr": 1.18, "cortex_mean": 298.0, "medulla_mean": 252.5},
    "flag": {"left": "plausible", "right": "compartments_likely_not_separated"},
  },
  "verdict": "INFO | WARN | UNKNOWN | N/A",
  "reason": "right kidney CMR 1.18 – cortex and medulla masks may not be separated",
}
```

how I plan to compute it (method):

1. Compute $CMR = \text{mean}(RBF[\text{cortex}]) / \text{mean}(RBF[\text{medulla}])$ per kidney. Guard `medulla_mean ≈ 0`.
2. Also emit both compartment means, so a reader can see *which* side of the ratio is anomalous.
3. Interpret **only in the low direction**, and only as evidence about segmentation.
4. Emit **INFO** normally; emit **WARN** only in the low-ratio segmentation-failure case. **Never FAIL**.
5. Excluded from overall aggregation by default (INFO-class, like brain check 8.2), so an uncalibrated ratio cannot silently drag a scan's verdict.

PASS / WARN / FAIL: | outcome | condition | |---|---| INFO | CMR ≥ ~1.5 — reported, no quality claim made | WARN | CMR < ~1.5 → "cortex and medulla masks may not be separated" (a mask problem, not a kidney problem) | FAIL | **never** | N/A | no medulla mask and none derivable — the consensus default protocol produces cortex ROIs only |

thresholds & sources:

- **The ~1.5 trip point** — **UNCALIBRATED**. No paper states a cortex:medulla ratio threshold for any purpose. It is placed between two empirically separated clusters, with margin:

STUDY	CMR (DERIVED)	SEGMENTATION STATUS
hammon2016_reproducibility_15t_semiautomatic.pdf	337.10 / 279.61 = 1.21	authors state: "the segmentation of the medulla may contain parts of the cortex what explains higher than expected medullary perfusion values"

STUDY	CMR (DERIVED)	SEGMENTATION STATUS
shirvani2019_motioncorrected_multiparametric_renal_asl.pdf	184.84 / 168.49 = 1.10	3D-GRASE; same paper reports cortex T1 799.61 ms vs medulla 807.11 ms at 3 T — implausibly low and essentially identical, against Gillis's medulla T1 of 1651 ± 86 ms at 3 T. Their T1-threshold segmentation almost certainly did not separate the compartments
garciarui2025 (3 T)	310.63 / 133.83 = 2.32	clean
garciarui2025 (1.5 T)	347.65 / 153.64 = 2.26	clean
hartevel2020 (FAIR)	362 / 140 = 2.59	clean
hartevel2020 (pCASL)	201 / 84 = 2.39	clean
haddock2019_furosemide_cortex_medulla.pdf	273 / 38 = 7.2	strict <i>inner-half-of-medulla</i> ROI
XCAT phantom	250 / 50 = 5.0	input assumption , cites Roberts 1995 — cannot validate itself

- 📄 **Physiological backing for the low-direction reading:** only ~10% of renal blood flow perfuses the medulla (📄 [odudu2018](#)). A CMR near 1 is anatomically impossible, so it is evidence the two masks are drawing from the same tissue.
- 📄 **Why the high direction carries no verdict:** high CMR (5–8.5) simply reflects a strict inner-medulla ROI convention (Haddock; Artz). Convention, not defect. Never flag it.

⚠️ **Three hard limits on what CMR may be used for — all of which must appear in the report text.**

(1) **It is not a disease marker, and it is not even monotone with disease.** Across the two severity-staged cohorts it moves in **opposite directions**: [zhang2024_ckd_oxygenation_perfusion.pdf](#) rises 1.57 (healthy) → 2.01 (CKD5), while [shi2026_membranous_nephropathy.pdf](#) falls 2.08 (healthy) → 1.75 (moderate-severe). A quantity that moves both ways with disease cannot discriminate disease.

(2) **Its denominator is the least reliable measurement in renal ASL.** Consensus R10.2 (89%) says medullary RBF is not reliable; medullary inter-visit ICC spans 0.13–0.96 with CV up to 37% (📄 [odudu2018](#)); Garcia-Ruiz measured medullary inter-session ICC of **0.08 at 3 T**. [gillis2014](#) measured medullary T1 but declined to quantify medullary perfusion at all, citing heterogeneity from large vessels and the renal pelvis. **CMR inherits all of that.** That is precisely why it may flag the masks and may never grade the kidney.

(3) **The clean cluster is not as tight as it looks.** The four "clean" studies above happen to be the ones whose *absolute cortical values* already agree closely (310–362), so selecting on medulla-reporting partly selects on agreement. Treat the 2.26–2.65 cluster as suggestive, not as a reference interval.

🔗 **needs:** cortex **and** medulla masks per kidney. Without a medulla mask → N/A (does not escalate).

🔗 **catches:** cortex/medulla masks that are not actually separated — the dominant renal segmentation failure, and one with a published worked example (Hammon's own admission). It also catches a global scaling error only in the sense that it *cannot* be fooled by one, which is why it is worth having alongside the absolute level check K3.1.

K3.3 Left-vs-right consistency • REQUIRED

🎯 **what it checks:** do the two kidneys give consistent cortical RBF? A large gap is a flag for human review — it may be genuine unilateral pathology (renal artery stenosis, a single functioning kidney) or it may be that one kidney's masks or motion correction failed.

📄 **inputs:** `cortical_rbf` per kidney from K3.1.

📄 **output:** `{"asymmetry_index": 0.07, "left": 312.0, "right": 298.0}` with a WARN above the tolerance.

🔧 **method:** $AI = (L - R) / (0.5 \cdot (L + R))$ — 📄 the same normalised-difference definition as the brain asymmetry check.

📊 **PASS / WARN / FAIL:** | outcome | condition | |
 PASS | $|AI| \leq \sim 0.20$ | |
 WARN | $|AI| > \sim 0.20$ → flag for human review (may be real pathology) | |
 FAIL | **never** — asymmetry is a review flag, not a rejection | |
 N/A | single kidney (transplant, nephrectomy, agenesis) |

📌 **thresholds & sources:**

- 📄 **No significant left-right perfusion difference** in four independent cohorts: cortex $P = 0.93$ (FAIR) and $P = 0.52$ (pCASL) with within-subject L–R correlation $r = 0.83 / 0.94$ (📄 [hartevel2020](#)); $P > 0.05$ in $n=30$ (Li 2020); $p = 0.93$ in $n=12$ (📄 [gillis2014](#)); non-significant across three sessions in $n=10$ PET/MR (📄 [olsen2025](#)).
- 📄 **But a consistent small left > right bias exists** and must be tolerated: ~0.5–6.8% in four Table-S1 studies (Boss 336/316, Cai 193/189, Rapacchi 315/295, Ren 392/390 — left higher in **four of four**), and ~9–13% across three PET/MR sessions (📄 [olsen2025](#)). A tolerance tighter than ~13% would fire on normal anatomy.
- **The ~0.20 tolerance** — 🔧 **UNCALIBRATED**. Set above the largest observed normal bias (~13%) with margin, and comfortably above the cortical between-visit CV of 4–13%.

- ⚠️ **Put this check on perfusion, not on transit time.** Arterial transit time *does* differ significantly between kidneys with FAIR (cortex 0.50 ± 0.13 vs 0.45 ± 0.14 s, $P = 0.010$) even when perfusion does not ([hartevelde2020](#)).

🐛 **catches:** one-sided mask or registration failure; genuinely asymmetric pathology surfaced for review.

Module K4 — Masks & co-registration

K4.1 Per-kidney mask integrity · REQUIRED

🔍 **what it checks:** are the supplied masks structurally sane — two separate connected components, non-empty, non-overlapping, inside the FOV, and large enough to support a cortical mean?

📁 **inputs:** `kidney_masks`, `cortex_masks`, `medulla_masks` (optional), `voxel_size_mm`.

📁 **output:**

```
{
  "metric": {
    "left": {"n_voxels": 812, "n_components": 1, "touches_fov_edge": False,
             "cortex_voxels": 430, "cortex_thickness_vox_est": 2.1},
    "right": {"n_voxels": 790, "n_components": 3, "touches_fov_edge": True, ...},
    "masks_overlap": False,
  },
  "verdict": "PASS | WARN | FAIL | UNKNOWN",
  "reason": "right kidney mask is fragmented (3 components) and touches the FOV edge",
}
```

🔧 **method:** pure-NumPy connected-component labelling; voxel counts; boundary contact test; cortex $\cap$ medulla overlap test; estimate cortical thickness in voxels as `cortex_volume / cortex_surface`.

📊 **PASS / WARN / FAIL:** | outcome | condition | |---|---| | ✅ PASS | one component per kidney, no overlap, not clipped, $\geq \sim 50$ cortical voxels | | ⚠️ WARN | fragmented, clipped by FOV, or estimated cortical thickness ≤ 2 voxels | | ❌ FAIL | empty mask, or cortex and medulla masks overlap | | ? UNKNOWN | no masks supplied |

📌 **thresholds & sources:**

- **Cortical-thickness-to-voxel-size warning** — 🔧 **UNCALIBRATED**, but the *mechanism* is published: 📄 consensus, "*In the case of advanced disease, the reduction in cortical thickness can be severe enough such that it approaches the typical dimensions of the ASL voxels ... The mixing of perfusion signal from the cortex with medullary signal may bias cortical RBF estimates to lower values ... cortical RBF results from ASL should be interpreted with caution in cases where cortical thinning is evident.*" With recommended in-plane resolution of 2–4 mm (R6.10), a thinned CKD cortex is genuinely only ~ 1 –2 voxels wide. **This means a low cortical RBF can be an artefact of anatomy rather than a perfusion finding** — which is a strong reason not to set a low-side FAIL in K3.1.
- **≥ 50 cortical voxels** — 🔧 **UNCALIBRATED**. For context, [shi2026](#) used one cortical ROI of 70–110 voxels and three medullary ROIs of 12–25 voxels each.
- **All mask-structure rules** — 📖 definitions.
- 📄 [bones2021_vsas1_label_dynamics_kidney.pdf](#) erodes the kidney mask with a 2×2 kernel to suppress partial-volume contamination — offered as an optional pre-step.

K4.2 ASL ↔ M0 and per-kidney registration · REQUIRED

🔍 **what it checks:** are the perfusion series, the M0 and the masks actually in the same space, and was registration done **per kidney**?

📁 **inputs:** affines and shapes of `delta_m_4d` / `m0` / masks; optionally the registration transforms.

📊 **PASS / WARN / FAIL:** | outcome | condition | |---|---| | ✅ PASS | grids match (or transforms supplied); per-kidney registration declared | | ⚠️ WARN | a **single global rigid transform** was used for both kidneys | | ❌ FAIL | grid/affine mismatch with no transform — every ROI statistic would be meaningless | | ? UNKNOWN | no registration provenance recorded |

📌 **thresholds & sources:**

- 📄 **Per-kidney registration is a consensus requirement**, R8.1 (13% abstentions, **100% agreement**), with the explicit note: "*the kidneys should be registered separately if using rigid/affine transformations as they move independently.*" Corroborated by three independent implementations: per-kidney translation-only Euler in Elastix ([bones2019](#)), per-subject/per-kidney/ **per-slice** with a PCA groupwise metric ([bones2022](#)), and per-repetition-and-kidney rigid affine ([olsen2025](#)). **This has no brain analogue at all** — the brain is one rigid body.
- ⚠️ 📄 **Heavy background suppression can break ASL-driven registration to M0.** Registration success was **54%** using the BGS ASL images themselves versus **100%** using separately-acquired fat images ([bones2019](#) , $n=9$, heaviest suppression). So BS quality and registration quality trade off, and the toolbox must check them separately rather than assuming good BS implies good alignment.

K4.3 Slice coverage / usable-slice fraction · REQUIRED

🔍 **what it checks:** how many acquired slices survived to contribute to the reported value.

 **output:** {"n_slices_acquired": 5, "n_slices_used": 3, "usable_fraction": 0.6, "edge_slices_dropped": 2}

 **method:** count slices with a non-empty mask and finite perfusion values; report separately for first/last slices of the stack.

 **PASS / WARN / FAIL:**  **UNCALIBRATED throughout** — report the fraction, WARN below ~0.5.

 **sources (context, not thresholds):**

-  In a 5-slice FAIR protocol, only **48.3%** of 60 single-kidney scans retained all five slices; the full distribution was 1.7% / 8.3% / 16.7% / 25.0% / 48.3% for one to five usable slices ([olsen2025](#)). Slices were dropped for motion-compensation failure and localised artefact — by **visual** assessment, not by any metric.
 -  **Edge slices fail systematically:** *"maps with significant noise were excluded (mostly generated from slices at the very beginning or end of the imaging stack, mainly due to partial volume effect with the nearby structures) ... The middle slice ... was the most representative one"* ([radovic2022_paediatric_allograft_asl.pdf](#)). So weight the check by slice position.
 -  Coverage is protocol-conditional: consensus R6.1 (95%) makes **2D single-slice the default**, so "whole-kidney coverage" must never be assumed. Gate this check on readout scheme.
-

● STREAM A — QC of the raw data (Modules K5–K8)

● Module K5 — Schema, data type & control-label

K5.1 Schema / metadata completeness · REQUIRED

🔗 **what it checks:** is enough acquisition metadata present to run Module K8?

⚠️ **BIDS does not cover the kidney.** The ASL-BIDS authors say so in print: "*Whereas ASL-BIDS could perhaps be used for other body parts, ASL-BIDS is validated in ASL images of the brain only*" ([clement2022_asl_bids.pdf](#)) — and their example of an out-of-scope application is a citation to the renal ASL consensus. I verified the spec directly: `kidney`, `renal` and `abdom` occur **zero** times in BIDS v1.11.1 (schema 1.2.1), against 64 occurrences of `brain`. There is no registered BEP for body/abdominal/renal imaging among all 47 BEPs, and zero issues mentioning kidney/renal on the bids-specification tracker. (*One open proposal does exist for generic body-part tagging — bids-specification issue #1569, "BIDS-BodyPart extracted from BEP025", filed by the BEP025 lead — so the door is not closed, it is just not renal.*)

Therefore the schema check is anchored on the renal consensus, not on BIDS. The published target is Nery 2020 **Table 4, "Minimum set of parameters to be reported in ASL studies"**, agreed at 81–100%:

- **General MR (19 items):** scanner manufacturer/model, receive coil type, pixel bandwidth, fat suppression, field of view, magnetic field strength, flip angle, image orientation, in-plane resolution, number of slices, parallel imaging (technique + acceleration factor), partial Fourier, physiological triggering/gating, readout pulse sequence type, slice gap, slice ordering, slice thickness, echo time, repetition time.
- **ASL-specific (6 items):** background suppression, inflow time(s)/PLD(s), labelling duration, labelling type, number of averages, quantification model.

Of those ~25 items, roughly 8 have no BIDS field at all — most importantly **physiological triggering/gating**, **fat suppression**, **pixel bandwidth** and **slice gap**. Two further renal essentials have no BIDS field either: **organ/laterality** (which kidney) and **native vs transplant**.

📊 **PASS / WARN / FAIL:** | outcome | condition | |---|---| | ✅ PASS | all Table-4 items resolvable from the sidecar/header | | ⚠️ WARN | items missing (list them); K8 checks for those items → UNKNOWN | | ❓ UNKNOWN | no metadata at all |

📌 **thresholds & sources:**

- **Table 4 item list** — 📄 [nery2020_renal_asl_consensus.pdf](#), agreed 81–100%. **PUBLISHED**, and it is a presence test, so it needs no calibration — the highest-confidence check in this document.
- **BIDS hooks that do exist:** `BodyPart` (optional, free-text, DICOM 0018,0015), plus `BodyPartDetails` and `BodyPartDetailsOntology` (also optional). There is **no filename entity** to encode organ for ASL — the `voi-<label>` entity is available to MRS only, and `perf` entities are limited to sub/ses/acq/run/rec/dir/part/echo. So kidney/laterality must be a clearly-labelled **non-standard extension**, and should be flagged as such in the report rather than presented as BIDS-compliant.
- ⚠️ **Two BIDS traps to encode in the reader:**
 1. BIDS `SliceThickness` is the **microscopy** field, defined in micrometres. Do **not** map Nery's "slice thickness" onto it — use `AcquisitionVoxelSize` (mm, explicitly excluding inter-slice gaps).
 2. `LabelingSlabThickness` says "*For non-selective FAIR a zero is entered*" while the schema sets `exclusiveMinimum: 0`. I traced it into `bids-validator/src/schema/applyRules.ts`, which passes the raw metadata object into ajv, so entering the prose-instructed `0` raises a validation error. **FAIR is the dominant renal PASL variant, so renal data hits this.** No BIDS issue mentions the field — this is an unreported bug and a good, cheap first upstream contribution.

K5.2 Data-type / geometry detection · REQUIRED (routing, INFO)

🔗 **what it checks:** classifies the renal acquisition so the right checks run — the renal analogue of brain check 8.2, and equally load-bearing given how little renal metadata exists in practice.

Axes to classify: labelling (FAIR/PASL vs PCASL — consensus R3.1, 93%, says **both are adequate**, so neither may be assumed); single-TI vs multi-TI/PLD; readout (2D single-slice / 2D multislice / 3D); orientation (coronal-oblique vs sagittal-oblique vs axial); native vs transplant; M0 present; background suppression; laterality present.

📊 **verdict:** **INFO** — routing only, excluded from aggregation.

📌 **sources:**

- 📄 Both labelling schemes endorsed (R3.1, 6% abst / 93%); single time point recommended (R3.2, 86%); 1.5 T and 3 T both adequate (R2.1, 87%).
- 📄 The routing consequences are deterministic and matter a great deal: the **20-pair minimum is scoped to 2D readouts** (see K8.2), the **QUIPSS II requirement applies only to FAIR** (K8.1), the **labelling-plane geometry check applies only to PCASL** (K8.3), and **transplant geometry inverts the orientation check** (K8.4).

K5.3 Control vs label ordering · REQUIRED

🔗 **what it checks:** the same catastrophic-swap test as brain — is `mean(control) > mean(label)` ?

 **method:** identical in form to brain 5.3. **Different in reliability,** for three renal reasons:

1. **The margin is larger than brain but the noise is much larger too.** Renal cortical PWS is ~3% of M0 (K2.2) versus ~1% in brain — so the *signal* is bigger. But renal acquisitions are **free breathing** by consensus (R7.4, 76%), so volume-to-volume intensity swings from the kidney moving through the slice can easily exceed a 3% label difference. The brain check assumes a stationary organ; kidney does not satisfy that.
2. **Therefore compute it on a motion-robust statistic.** Use the **median across repetitions of the within-kidney-mask mean**, not the grand mean of all volumes, and report the fraction of pairs in which control > label rather than only the pooled comparison. A swap should show as a *consistent* sign flip across nearly all pairs, not a small mean difference.
3. **Background suppression is recommended for renal ASL** (R7.2, 5% abst / **80%** — among the lower-agreement acquisition statements; it ties with R5.1, and only R6.4 at 75% and R7.4 at 76% sit below it), so the swap test will frequently be N/A on renal data for the same reason as brain.

 **PASS / WARN / FAIL:** | outcome | condition | | PASS | control > label in a clear majority of pairs | | FAIL | label > control consistently → subtraction runs backwards | | WARN | inconsistent sign across pairs → suggests motion dominating the label difference | | N/A | background suppression ON, or pre-subtracted ΔM supplied |

 **sources:**  physics (control retains full tissue signal);  the ~5%-of-control renal PWS magnitude from [odudu2018](#) and the ~3% measured value from [garciarui2025](#). **No published renal swap threshold exists** —  the "clear majority of pairs" rule is UNCALIBRATED.

Module K6 — M0 calibration

This is the module that transfers best from brain, and the reason is worth stating: the renal consensus imposes the **same M0 rule as the ASL White Paper, not a contrasting one**. An earlier reading of the evidence suggested renal BS rules were "opposite polarity" to brain — that is wrong, and the correction matters because acting on it would have inverted a working check.

K6.1 M0 present · **REQUIRED**

 **sources:**  R9.1 (0% abstentions, **94% agreement**): "M0 acquisition is mandatory", and in the body: "we consider acquisition of a separate PD image (also referred to as M0 image) a mandatory step for ASL quantification." **This is stronger wording than the brain White Paper**, which only "recommends" a separately acquired PD image, and BIDS explicitly permits **M0Type: "Absent"**. So a missing M0 may legitimately be a **harder failure in kidney than in brain** — one of the very few places where a renal check can be *stricter* than its brain counterpart with a published warrant.

 **verdict:** FAIL if absent and quantified RBF is claimed; WARN if absent and only ΔM is supplied.

K6.2 M0 acquired without labelling and without background suppression · **REQUIRED**

 **sources:**  consensus, verbatim: "The PD image should be acquired without labelling or BGS and using a similar readout and acquisition parameters, with the exception that a long TR should be used." **Identical rule to brain.** Ports unchanged.  Note R7.2 recommends BS *for the ASL pairs* — that is not a contradiction, it is the same split as brain.

K6.3 M0 TR ≥ 5 s (or corrected) · **REQUIRED**

 **sources:**  consensus, verbatim: "If this image is acquired without waiting for a sufficiently (> 5 s) long recovery time (TR), the SI_PD should be corrected for incomplete relaxation using the equation: $SI_PD,corr = SI_PD / (1 - \exp(-TR/T1,tissue))$, where $T1,tissue$ is an estimate of the kidney $T1$." Same 5 s threshold and same functional form as the brain White Paper — **PUBLISHED and directly portable**.

 **But the correction cannot actually be applied, because the consensus never gives a number.** I searched the full text: it contains **no numeric kidney T1 anywhere**. And kidney has **two** tissue T1s that differ by ~400 ms, so a single value is a real approximation:

	1.5 T	3 T
cortex	~827–1080 ms	~1124–1406 ms
medulla	~1054–1428 ms	~1388–1685 ms

( [wolf2018_renal_t1_t2_systematic_review.pdf](#), healthy-subject ranges across studies.) The OSIP living lexicon lists cortex 966 ± 58 ms at 1.5 T and 1142 ± 154 ms at 3 T — but that 3 T value is ~16% below [garciarui2025](#)'s direct measurement of 1356.02 ms, so it is not a safe default.

Proposed behaviour, pending mentor input (open question 4): if $TR < 5$ s and no T1 map is supplied, **do not silently correct** — emit WARN with the magnitude of the correction under both a cortex and a medulla T1, so the user sees the size of the uncertainty. If a T1 map is supplied, apply the correction per-compartment.  the choice of default T1 is UNCALIBRATED.

K6.4 M0 geometry match · **STRETCH**

Same as brain: does M0 share the ASL grid? Consensus adds a specific note worth encoding —  "Smoothing the M0 data may reduce the impact of misregistration on perfusion quantification" — which implies a dedicated ASL↔M0 registration check is warranted (it is K4.2).

Module K7 — Respiratory motion & outlier rejection ★

This is the module with no brain equivalent, and the one the mentors specifically asked to see.

The consensus is unambiguous that this is *the* renal problem: "Subject breathing induces kidney displacements up to an order of magnitude larger than the typical ASL voxel size, which if unaccounted for cause a significant loss of image quality in ASL." It then recommends registration (R8.1, 100%) and outlier rejection (R8.2, 0% abstentions, 100% agreement) — and specifies, for both, **no method and no threshold whatsoever**. That gap is the clearest, best-evidenced opening for this toolbox in the entire renal literature.

📄 An honest quantification of "order of magnitude", because the consensus rhetoric overstates its own numbers. Measured 4D-CT kidney excursion is 11.1 ± 4.8 mm cranio-caudal (range 2.5–20.5), 3.6 ± 2.1 mm AP, 1.7 ± 1.4 mm RL across 39 kidneys (yamashita2014_renal_motion_4dct.pdf ; corroborated by siva2013_kidney_motion_4dct.pdf , which tracked the kidney apex rather than the centre of gravity and found 7.4–7.5 mm). Against the consensus's own recommended 2–4 mm in-plane resolution, 11 mm is **~3–5 voxels**, not 10–20. Use the measured figure, not the rhetorical one. The important part is the **anisotropy**: CC:AP:RL $\approx 6.5 : 2 : 1$.

K7.1 Per-kidney respiratory displacement · REQUIRED

🔍 what it checks: how far each kidney moved, along which axis, during the series.

📁 inputs:

```
{
  "delta_m_4d": "NIfTI 4D float",          # or the raw control/label series
  "kidney_masks": {"left": ..., "right": ...},
  "voxel_size_mm": [3.0, 3.0, 5.0],
  "affine": "NIfTI affine",               # to resolve which axis is cranio-caudal
  "transforms": "optional per-kidney per-volume transforms, if registration already ran",
}
```

📁 output:

```
{
  "metric": {
    "left": {
      "median_interframe_cc_mm": 2.1, "max_cc_excursion_mm": 9.4,
      "median_interframe_ap_mm": 0.8, "median_interframe_rl_mm": 0.4,
      "median_interframe_cc_vox": 0.70,          # the units that actually matter
      "through_plane_fraction": 0.18,           # share of displacement out of the imaging plane
    },
    "right": {...},
    "axis_convention": "CC resolved from affine, not assumed",
  },
  "verdict": "PASS | WARN | UNKNOWN",
  "reason": "left kidney median inter-frame CC displacement 0.70 voxels",
}
```

🔧 how I plan to compute it (method):

1. **Resolve anatomical axes from the affine** — never assume array-axis order. Coronal-oblique is the consensus-recommended orientation (R6.7, 93%), so the CC axis is usually in-plane, but this must be derived, not assumed.
2. For each kidney separately, compute the **intensity-weighted centroid** of the masked region in every volume (pure NumPy — no registration required, which keeps this runnable with zero heavy deps). If per-volume transforms are supplied, use their translations instead.
3. Project consecutive-volume displacements onto the CC / AP / RL axes; report **median and max** per axis, in **both mm and voxels**.
4. Compute a **through_plane_fraction** = share of total displacement normal to the imaging plane. This is the component registration **cannot** fix and is the documented cause of whole-subject data loss.
5. **Never collapse to one scalar**. An isotropic sum like brain FD would average away the CC axis, which carries ~6.5× the RL motion.

📊 **PASS / WARN / FAIL:** | outcome | condition | |--| | | PASS | median inter-frame CC displacement ≤ 1 voxel | | WARN | > 1 voxel, or **through_plane_fraction** high | | FAIL | **not on this metric alone** — motion severity is reported; the *consequence* is graded by K7.2 | | UNKNOWN | no 4D series or no mask |

📌 thresholds & sources — this is where the honesty matters most:

- There is **NO** published mm or voxel threshold for failing a renal ASL scan on motion. 🔧 The ≤ 1 -voxel WARN line is **UNCALIBRATED**. What exists, and what informed it:
 - 📄 cox2017_multiparametric_renal_mri_validation.pdf : "Individual perfusion weighted difference images (control-label) are calculated, inspected for motion (exclude > 1 voxel movement) or realigned" — the closest thing to a published rule. But it is **visual inspection**, it never states the voxel size, and it describes a group's general pipeline rather than a validated criterion.
 - 📄 Tan et al. 2014, *MRM* (2D navigator-gated renal FAIR at 3 T — **not in kidney/papers/** , **fetch before citing in a deliverable**): an **8 mm** navigator acceptance window, which is a *prospective gate* on a 1-D diaphragm navigator with 4 mm pixels, **not** a retrospective verdict threshold. It achieved residual

1.1 mm mean / 4 mm max SI displacement in 10 healthy volunteers — the only published quantification of residual renal motion after correction, and the nearest thing to an empirical "motion adequately handled" target.

- [robson2009](#) (bellows retrospective sorting, also not in [kidney/papers/](#)): keep roughly the **half** of acquisitions closest to end-expiration; reported perfusion CoV fell from 6.6% (timed breathing) to 1.7% with end-expiration sorting.
- [song2017_respiratory_motion_prediction.pdf](#): a **±1 mm** target prediction accuracy — an engineering target for a prospective controller, not a quality tolerance.
- **FD and DVARS do not transfer.** I ran the search: `"framewise displacement" AND (kidney OR "renal perfusion") AND MRI NOT brain` returns **0 hits** on Europe PMC; every FD/DVARS + ASL + renal hit is a brain study in renal patients. Beyond the absence, there are three structural reasons FD is the wrong shape for kidney: motion is strongly **anisotropic**, the two kidneys move **independently** (so a single 6-DOF series is ill-defined), and the brain FD formula converts rotations using a **50 mm head-radius sphere** which is meaningless for an abdominal organ.
- Through-plane motion is a documented cause of total data loss: "One subject was excluded because of excessive through-plane motion during the scans" (1 of 6, [franklin2021](#)); and "One subject indicated difficulties with the paced-breathing protocol, resulting in major kidney displacements with in-plane and through-plane motion corrupting the PWIs" (1 of 10, [bones2019](#)).

K7.2 Subtraction-outlier rate · REQUIRED ★ (the one genuinely implementable published rule family)

what it checks: how many control-label pairs are corrupted enough to reject — the direct operationalisation of consensus R8.2.

inputs: `delta_m_4d`, `kidney_masks`, `rule` (config), plus the rule's parameters.

output:

```
{
  "metric": {
    "rule": "hartevelde_2sd_20pct",
    "left": {"n_pairs": 25, "n_rejected": 2, "rejected_fraction": 0.08,
            "surviving_pairs": 23},
    "right": {"n_pairs": 25, "n_rejected": 6, "rejected_fraction": 0.24,
             "surviving_pairs": 19},
    "note": "rule fires in ~2/3 of NORMAL healthy datasets – count, not occurrence, is the signal",
  },
  "verdict": "PASS | WARN | FAIL | UNKNOWN",
  "reason": "right kidney: 6 of 25 pairs rejected; surviving pairs (19) below the 20-pair minimum",
}
```

how I plan to compute it (method):

1. Build the voxel-wise temporal mean and SD of ΔM across repetitions, inside each kidney mask.
2. For each pair, compute the fraction of within-mask voxels deviating by more than k -SD.
3. Reject the pair if that fraction exceeds the rule's limit.
4. Report `n_rejected`, `rejected_fraction`, and `surviving_pairs` — the last of which feeds straight into K8.2's consensus minimum, which is the closest thing to a principled FAIL available.

Three published parameterisations, all shipped, `hartevelde_2sd_20pct` as default:

RULE ID	CRITERION	PROVENANCE	OBSERVED BEHAVIOUR
<code>hartevelde_2sd_20pct</code>	reject pair if > 20% of kidney voxels deviate > ±2 SD from the voxel-wise temporal mean	hartevelde2020_multidelay_fair_vs_pcasl.pdf , 3 T, n=15–16	fired in 18/27 FAIR and 19/27 pCASL datasets, max 2 excluded pairs per delay
<code>bones_80pct_2sd</code>	keep pair only if > 80% of kidney voxels lie within ±2 SD (same rule, stated positively)	bones2021_vsasl_label_dynamics_kidney.pdf , 1.5 T, n=7 — authors call it "an empirically chosen threshold"	—
<code>hartevelde_paed_1p5sd_20pct</code>	reject if > 20% of voxels deviate > ±1.5 SD	hartevelde2022_paediatric_neuroblastoma_nephroblastoma_pcasl.pdf , 1.5 T, n=10 children	1 (range 0–3) of 10 pairs rejected on average
<code>garciarui2_2sd_plus_rmse</code>	±2 SD on the time series, plus discard slices whose multi-delay model-fit RMSE exceeds mean + 2 SD	garciarui2025 , 1.5/3 T, n=16	1% of slices excluded; no participant lost entirely

PASS / WARN / FAIL: | outcome | condition | |---|---| | **PASS** | ≤ 2 pairs rejected per kidney **and** surviving pairs ≥ the applicable consensus minimum | | **WARN** | 3+ pairs rejected, or the rejected fraction is much higher in one kidney than the other | | **FAIL** | surviving pairs < consensus minimum (2D readouts only) — see K8.2) | | **UNKNOWN** | no 4D series or no mask |

📌 thresholds & sources:

- **The rules themselves** — **IMPLEMENTATION**. Each is one group's preprocessing choice, verbatim quotable, never validated as a quality criterion.
- **The "≤ 2 pairs" PASS line** — **UNCALIBRATED**, but it is the most defensible uncalibrated number in this document, because it is anchored on an *observed ceiling in known-good data*: Hartevelde's rule fired in roughly two-thirds of healthy datasets with a maximum of two rejected pairs. **The critical calibration note, which must appear in the report: the rule firing is not itself a defect signal** — it fires on most normal data. Only the count matters.
- **The FAIL condition is the only one with published backing**, and even that is indirect: it is R4.7/R5.12's 20-pair minimum applied to the *surviving* pairs. Whether that is a fair use of the recommendation is **open question 3**.
- The consensus's own description of the state of practice: "Outlier rejection methods ... have relied on manual or automatic approaches, including using data from external sensors such as respiratory bellows." No standard algorithm exists.

K7.3 Breathing strategy & gating efficiency • REQUIRED (metadata) / STRETCH (trace)

what it checks: was a consensus-conformant breathing strategy used, and if a respiratory trace exists, how much data survived gating?

PASS / WARN / FAIL: | outcome | condition | |---|---| | **PASS** | free breathing, or free breathing with respiratory triggering | | **WARN** | **breath-hold** — R7.3 (0% abst, **94%**): "Breath-hold scans are not recommended for clinical renal ASL" | | **INFO** | gating efficiency reported if a trace is present | | **UNKNOWN** | breathing strategy not recorded (and it has **no BIDS field**) |

📌 thresholds & sources:

- R7.4 (0% abst, **76%**): free breathing preferred. **This clears the consensus bar (75%) by one point** and is the second-weakest statement this design relies on, after R6.4's 75% — so free-breathing conformance should never be more than a WARN.
- R7.5 (5% abst, 95%): respiratory triggering advantageous, at the cost of scan time.
- **Gating efficiency: report only, NO threshold.** Reference values are **50 ± 13%** (coronal navigator) and **41 ± 9%** (sagittal) in 10 healthy volunteers (Tan 2014). **A "below 40–50% indicates poor compliance" rule would be wrong:** 41 ± 9% is the normal sagittal mean, so such a floor would fail roughly half of normal scans. This is exactly the cohort-mean-as-threshold error and is called out here so it does not get re-introduced later.
- **An external respiratory trace is not sufficient on its own.** [bones2020_vsasl_labeling_respiratory_motion.pdf](#) found **no one-to-one correlation** between bellows signal change and spurious labelling at the individual-repetition level: "Some repetitions had tissue labeling without the bellows indicating motion, others showed no spurious labeling even when motion was detected." Image-derived metrics (K7.1, K7.2) are required regardless.
- **Where the trace legally lives:** BIDS [timeseries__mri_no_task](#) permits [*_physio.tsv.gz](#) in the [perf](#) datatype, and [respiratory](#) is a defined column — so [perf/sub-01_recording-respiratory_physio.tsv.gz](#) is spec-legal. This is one of the few places renal ASL data *does* have a standards-compliant home.

K7.4 Spurious-labelling signature • STRETCH

what it checks: a concrete, automatable artefact pattern with a published description — **homogeneously elevated ΔM across the entire kidney**, i.e. loss of cortico-medullary contrast combined with an abnormally high whole-kidney mean.

📌 **sources:** [bones2020_vsasl_labeling_respiratory_motion.pdf](#) : a pair "was judged to contain spurious labeling when it showed homogeneously high ΔM over the entire kidney ROI"; **25.4%** of repetitions affected with foot-head labelling gradients, falling to 4.3% (RL) and 4.9% (AP), with **5 of 15 subjects excluded**. Detection thresholds UNCALIBRATED. Caveat: this is a velocity-selective-ASL-specific mechanism, so the check should be gated on labelling scheme — but the *signature* (flat, bright, contrast-free kidney) is the most concretely describable renal artefact pattern in the literature and is worth implementing.

◆ Module K8 — Protocol conformance vs the renal consensus ★

This is the richest PUBLISHED block in the design. Every threshold here is a numbered consensus statement with an agreement percentage. Each is a straight metadata range test — no calibration needed, and no judgement smuggled in.

K8.1 FAIR / PASL timing conformance • REQUIRED (gated on labelling = PASL)

PARAMETER	RECOMMENDED	STATEMENT	ABST / AGREE
inversion time TI (single-TI)	1.8–2.0 s	R4.4	10% / 89%
bolus duration TI1	1.0–1.2 s	R4.6	25% / 92%
bolus-width control (QUIPSS II / Q2TIPS)	mandatory for quantification	R4.5	25% / 100%
selective slab excludes the aorta	required	R4.2	6% / 100%
selective slab thickness	imaging slab + 10–20 mm	R4.3	13% / 86%
FOCI pulse for selective inversion	recommended	R4.1	19% / 92%

📌 **PUBLISHED** from [nery2020_renal_asl_consensus.pdf](#) .

⚠️ **The QUIPSS II rule needs a design decision (open question 8).** The consensus wording is hard — "an approach for controlling the temporal width of the bolus (QUIPSS II or Q2TIPS) **must** be used to quantify perfusion" — but the same paper concedes that "many studies of renal perfusion with pulsed ASL have not included this saturation. Quantification is typically performed by replacing T1 with T1 in the RBF equation ... This approach maximizes signal but **will underestimate flow by a transit time-dependent factor**." So a hard FAIL would reject a large slice of the published renal literature. **Proposed: WARN, with the bias direction stated in the reason string** (flow underestimated, magnitude unknown). Note also the 25% abstention — a quarter of the panel declined to vote. This is *softer* than the brain White Paper, which says single-T1 PASL without QUIPSS II "cannot be reliably converted into CBF" — a rare case where the renal rule is looser than the brain one.

K8.2 PCASL timing & averaging conformance · REQUIRED (gated on labelling = PCASL)

PARAMETER	RECOMMENDED	STATEMENT	ABST / AGREE
labelling duration τ	1.5–1.8 s	R5.2	10% / 100%
post-labelling delay (single-PLD)	1.2–1.5 s	R5.11	19% / 100%
minimum ASL pairs	≥ 20	R5.12 (PCASL) / R4.7 (FAIR)	14% / 83% · 10% / 89%
TR (labelling + readout)	4–6 s	R6.12	0% / 94%

⚠️ **The 20-pair rule is conditional on readout and must be gated.** The Table-2 statement reads "a minimum of 20 ASL pairs is recommended", but the Discussion scopes it: "The acquisition of a minimum of 20 ASL pairs (control and label images) **when using the recommended 2D readout is advised**." A blanket `n_pairs ≥ 20` check would mis-fire on 3D data — and would, for instance, fail [radovic2025_donor_recipient_paediatric_allograft.pdf](#), which used **6 pairs** on 3D-GRASE and explains why: 20 pairs would have taken 12 min 6 s instead of 3 min 42 s, "which was unacceptable in our (or any) study group." **Gate on MRAcquisitionType**.

Note also that PLD 1.2–1.5 s is **shorter** than the brain White Paper's adult 1.8–2.0 s — a constant that must not be inherited. And it is only marginally longer than the measured pCASL cortical ATT of 0.71 ± 0.25 s ([hartevel2020](#)), which is the physiological reason the recommendation sits where it does.

K8.3 Geometry & readout conformance · REQUIRED

PARAMETER	RECOMMENDED	STATEMENT	ABST / AGREE
slice orientation	coronal oblique, along the kidney major axis	R6.7	6% / 93%
default readout scheme	2D single-slice	R6.1	10% / 95%
preferred readout (2D single-slice)	spin-echo EPI	R6.4	5% / 75% ← lowest in the document
acceptable alternatives	bSSFP, single-shot RARE	R6.5	14% / 94%
3D as default	not recommended	R6.3	10% / 95%
slice thickness, 2D	4–8 mm	R6.8	19% / 100%
slice thickness, 3D	3–6 mm	R6.9	13% / 100%
in-plane resolution	2–4 mm	R6.10	0% / 93%
parallel-imaging acceleration	≤ R = 2	R6.11	19% / 100%
fat suppression	recommended	R7.6	5% / 90%
background suppression	recommended (2–4 inversion pulses typical)	R7.2	5% / 80%
PCASL labelling plane	⊥ to aorta, 8–10 cm superior to the highest kidney centre	R5.3 / R5.4	13% / 100% · 14% / 94%

📌 **PUBLISHED.** Two implementation notes:

- **Geometry is checkable from the NIFTI affine** (orientation, in-plane resolution, slice thickness) even with no sidecar — which matters, because renal data in the wild is as metadata-poor as the three brain test cases were. Fat suppression, pixel bandwidth, triggering and acceleration factor are **metadata-only** and will frequently be UNKNOWN.
- **R6.4 at 75% is a bare pass** (the consensus bar was 75%) and R6.11 is *permissive* wording ("may be used"), so both should WARN at most, never FAIL.
- ⚠️ **The rationale for coronal-oblique is itself QC-relevant and should go in the reason string:** 📄 "most kidney movement due to the respiratory cycle is within the image plane and so data can be corrected using image registration methods" — plus, for FAIR specifically, "this is necessary to avoid labelling of the inflowing vessels for the selective label." So a non-coronal-oblique FAIR acquisition is not just unconventional; it is a labelling-validity risk.

K8.4 Quantification-constant conformance · REQUIRED (if RBF is supplied)

CONSTANT	CONSENSUS VALUE	STATEMENT	ABST / AGREE	VS BRAIN
λ (tissue-blood partition)	0.9 mL/g	R9.4	5% / 90%	same number
T1 blood @ 3 T	1.65 s	R9.5	0% / 100%	same
T1 blood @ 1.5 T	1.48 s	R9.6	13% / 93%	differs (brain 1.35 s)
α , PASL	95%	R9.7	6% / 100%	differs (brain 0.98)
α , PCASL	85%	R9.8	13% / 86%	same
α correction per BGS pulse	$\times 0.93$ each	R9.9	19% / 100%	same mechanism
model	single-compartment, single-delay	R9.2	7% / 100%	same family

 all  **PUBLISHED**, with two caveats that belong in the report:

-  **$\lambda = 0.9$ is an admitted brain-borrowed placeholder.** Verbatim: "Since a reliable reference for the partition coefficient value in kidney was not known to this group, we recommend the use of a value of 0.9 mL/g, the average value for brain tissue." The panel adds that values "could be readily corrected when a more accurate value of λ is known." Tag it **PUBLISHED-as-consensus- assumption**, not PUBLISHED-as-measurement. For context, the reported literature spread is 0.80–0.94 in humans and a directly measured 0.91 in dog kidney ([hillaert2024_dog_fair_asl_lambda.pdf](#)), and [cox2017](#) uses 0.8.
-  **The α values are stated "neglecting background suppression loss", and BS is recommended.** So on the *recommended* renal protocol the raw 0.95/0.85 are wrong — they must be multiplied by 0.93 per BGS inversion pulse (e.g. PCASL with 3 pulses $\rightarrow 0.85 \times 0.93^3 = 0.684$). A toolbox that hard-codes 0.85 will be systematically wrong on exactly the protocol the consensus recommends. **The number of BGS pulses is rarely in metadata** — this is a real, checkable, frequently-violated conformance item.

Transplant profile · STRETCH

Transplant kidneys need a **geometry** profile, and — on current evidence — **no perfusion band**.

-  **Geometry genuinely differs and is checkable:** the allograft sits in the iliac fossa, so "*orienting the imaging slab along the long axis of the kidney was not always possible*" ([echeverriachasco2023_allograft_reproducibility.pdf](#)), and the consensus advises a labelling plane *perpendicular to the abdominal aorta, just above the bifurcation into the iliac arteries*, with PCASL "*may be preferable in the transplant setting*" (note: *may be*, and this is narrative text, not a numbered statement). Respiratory gating matters less — the lower abdomen moves less. So the K8.3 orientation check and K7.3 gating expectations should **invert** for transplants.
-  **No perfusion band.** The adult allograft literature is internally inconsistent: stable-function cortical RBF is 196.5 (3 T pCASL, [echeverriachasco2023](#)), 221.2 ([jiang2024](#)), 259.7 ([que2026_bold_asl_early_transplant.pdf](#)), 261.1 ([zhang2025](#)), 277.7 ([ma2025](#)) — a ~40% spread at nominally equivalent function, and two of those studies share identical acquisition parameters and eGFR strata, so they are probably not independent. And the direction of the transplant-vs-native offset is **not established**: the one ATT study reporting both found transplant cortex RBF *higher* (358.3 vs 271.8) with $n=4$ vs $n=7$ and a 17-year age gap running the wrong way.
-  **Do not reuse the published transplant cut-offs** (207.53, 210.06, 192.4, 88 mL/100 g/min). Those are **eGFR-discrimination operating points** — they answer "*is this graft functioning?*", a clinical question, not "*is this image good?*". Reusing them as QC bands would be a category error.

Paediatric profile · STRETCH (metadata-only in v1)

-  **No paediatric renal ASL normative data exists.** Four papers, ~49 children total, **none in healthy children, none in neonates, none reporting a cortex/medulla split**. The Pediatric Nephrology 2025 review states it outright: renal ASL is "*an experimental and not validated technique*" and "*establishing reference values ... in healthy pediatric controls is crucial, necessitating further research.*" Any paediatric renal band ships **UNCALIBRATED** or not at all.
-  **A frequently-miscited number, worth flagging so it does not enter the toolbox:** the widely quoted "295 mL/100 g/min" from [nery2019_paediatric_ckd_3dgrase_motion_correction.pdf](#) is from **five healthy ADULTS (age 32 ± 5)**, not children — the abstract says only "healthy volunteers". Its ±97 is a **voxelwise spatial SD**, not a between-subject SD; the between-subject spread is the range 245–343.
-  **What paediatric evidence *does* support** are metadata flags, not bands:  sedation or general anaesthesia may be needed under age 7 (and alters perfusion);  **14.3%** of paediatric transplant scans (3 of 21) were unusable for motion/cooperation;  motion correction moved inter-session WSCV from **0.506 → 0.201** and ICC from **0.372 → 0.833** in 11 children with severe CKD.
-  **The single best QC design lesson in the whole paediatric literature, and it is a warning about our own metrics:** in that same study, *intra-session* repeatability stayed **high on badly motion-corrupted data**, because "*both ASL runs shared the same M0 ... systematic errors because of a consistent misalignment between the M0 and the ASL data set in both ASL runs resulted in a repeatable mean RBF despite severe corruption in the RBF maps ...* **For this reason, intra-session WSCV and ICC alone should not be taken to be indicators of image quality directly.**" Repeatability must never be used as an image-quality proxy in this toolbox.

Multi-delay / arterial transit time · STRETCH

 Consensus R3.2 (86%) recommends **single time point** as the default; R3.3 (100%) says multi-delay is useful "*if delayed arrival time is suspected*"; R9.3 (95%) says two-compartment fitting is "*not recommended as the default*". And [odudu2018](#) says multi-delay "*cannot be currently universally recommended*". So ATT is out of scope for v1 — and should stay out:  [shimizu2017_att_corrected_rbf_pcasl.pdf](#)'s larger multi-PLD cohort concluded ATT "*has poor reproducibility and high variation*", with poor ICCs for cortical and medullary ATT alike. **A parameter that does not reproduce within a subject cannot anchor a threshold.**

Where a brain check CANNOT transfer, and why

The single most useful table in this document for a reviewer. ✔ = ports; ⚠ = ports with a changed constant or scope; ✖ = does not port.

BRAIN CHECK	KIDNEY	WHY
QE1 (Module 1)	✖	No cortex/medulla probability substrate; no consensus spRBF weighting (measured 2.3–2.6 vs phantom-assumed 5.0); human inter-observer cortex Dice ~0.77; medulla declared unreliable; no labelled renal dataset, so no curves and no cutoff.
QE1-Net	✖	Same, plus no training labels.
Spatial CoV, 0.67 cutoff	✖	Zero renal papers found (21 + 88 Europe PMC hits, all cerebral). A whole-kidney CoV would be dominated by the normal ~2.3–2.6× cortico-medullary gradient, i.e. by anatomy not artefact.
GM/WM ratio ≥ 1	⚠ → K3.2	Same scale-free <i>form</i> , opposite <i>role</i> . GM/WM < 1 is a published brain FAIL; CMR may only flag the masks , never the kidney, because its denominator is the least reliable measurement in renal ASL (ICC 0.08 at 3 T). And CMR is not monotone with disease — it moves in opposite directions across the two staged cohorts.
GM CBF 40–100 mL/100 g/min	✖	The renal equivalent (139–427) is a spread of study means across 53 heterogeneous studies whose healthy and patient ranges overlap almost entirely. And technical factors dominate: labelling ~1.8×, field strength ~11%, age ~20%, ROI convention ~7%.
Negative-GM fraction	✔ → K2.3	Ports directly; renal version uses the published two-sided 0–500 clip and cortex restriction.
FD / DVARS, 0.2–0.5 mm	✖ → K7.1	0 hits for framewise displacement in renal ASL. Kidney motion is anisotropic (CC:AP:RL ≈ 6.5:2:1), the kidneys move independently , and the 50 mm head-radius sphere is meaningless abdominally.
Motion correction = one rigid body	✖	Consensus R8.1: <i>"the kidneys should be registered separately ... as they move independently."</i> Per-kidney, sometimes per-slice.
Breath-hold irrelevant (brain is stationary)	✖ → K7.3	Free breathing is the consensus default (R7.4, 76%); breath-hold is explicitly not recommended (R7.3, 94%).
M0 present	⚠ ✔	Ports, and is stricter in kidney: <i>"M0 acquisition is mandatory"</i> (R9.1, 94%) vs "recommended" in the brain White Paper.
M0 without background suppression	✔	Identical rule. Explicitly <i>not</i> inverted in kidney — correcting an earlier misreading.
M0 TR ≥ 5 s	⚠	Same threshold and same correction formula — but T1, tissue is compartment-dependent (cortex vs medulla differ by ~400 ms) and the consensus gives no number .
$\lambda = 0.9$	⚠	Same number, materially weaker basis — the panel says outright it was borrowed from brain because no kidney reference was known.
T1 blood 1650 ms @ 3 T	✔	Same.
T1 blood 1350 ms @ 1.5 T	✖	Renal consensus says 1.48 s .
$\alpha = 0.98$ (PASL)	✖	Renal consensus says 95% . (PCASL 0.85 is unchanged.)
PLD 1.8–2.0 s (adult)	✖	Renal PCASL PLD is 1.2–1.5 s ; FAIR TI is 1.8–2.0 s but that is a different quantity.
PLD by age (White Paper Table 1)	✖	The renal consensus contains zero paediatric or age-specific recommendations among its 59 statements.
Control/label swap	⚠ → K5.3	Same physics, worse conditions: free breathing means volume-to-volume swings can exceed the ~3% label difference. Use a per-pair sign-consistency statistic, not a pooled mean.
BIDS schema check	✖ → K5.1	BIDS has zero kidney content; the only quantified-perfusion volume type is literally cbf = <i>cerebral</i> blood flow, defined by reference to the brain White Paper. Anchor on Nery Table 4 instead.
$\geq N$ control/label pairs	⚠ → K8.2	The renal 20-pair minimum is scoped to 2D readouts in the Discussion. Gate on MRAcquisitionType or it mis-fires on 3D.
Left-right asymmetry needs hemisphere masks	⚠ → K3.3	<i>Easier</i> in kidney — two separate organs, no midline problem — but a consistent ~0.5–13% left > right bias must be tolerated.
Default tissue priors (MNI fallback)	✖	No renal atlas space, no cortex/medulla priors. Masks must be supplied or the check is UNKNOWN.
5 mm FWHM pre-smoothing	✖	Tied to QE1's fitted constants, which do not exist for kidney — and it would blur across a cortex only ~1–2 voxels thick.

Inputs → possible outputs (the dependency map)

WHAT THE USER SUPPLIED	TSNR	PWS/M0	NEG / IMPLAUSIBLE	CORTICAL RBF	CMR	L-VS-R	MASK INTEGRITY	ASL↔M0 REG	COVERAGE	MOTION (K7.1)	OUTLIER RATE (K7.2)	PROTOCOL (K8)	M0 RULES (K6)
RBF map only	×	×	~ global only	×	×	×	×	×	×	×	×	×	×
+ whole-kidney masks (L/R)	×	×	✓	×	×	~ whole-kidney	✓	×	✓	×	×	×	×
+ cortex masks	×	×	✓	✓	×	✓	✓	×	✓	×	×	×	×
+ medulla masks / T1 map	×	×	✓	✓	✓	✓	✓	×	✓	×	×	×	×
4D ΔM series + masks	✓	×	✓	✓	✓	✓	✓	×	✓	✓	✓	×	×
+ M0	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	×	~ presence only
+ metadata (JSON / header)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
+ respiratory trace	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓ +	✓	✓	✓

Reading it:

- **The renal cliff is masks, not tissue maps.** In brain, a CBF map alone still yields a rough sCoV and histogram via an auto brain mask. In kidney, a map alone yields almost nothing, because there is no reliable auto-extraction and no atlas fallback.
- **Cortex masks are the highest-leverage single input** — they unlock the consensus-mandated reported quantity.
- **Metadata unlocks the entire published half of the design** (Module K8).
- + a respiratory trace refines K7.1 but does **not** replace it — bellows amplitude does not predict spurious labelling repetition-by-repetition.

? Open questions — for the mentors and for the renal specialist

These are the questions whose answers would actually change the design above.

Scope and ROI

1. **Cortex-only, or cortex + medulla?** R10.1 (100%) mandates cortical reporting and R10.2 (89%) declares medullary values unreliable — yet the CMR segmentation-integrity check (K3.2), which I think is the most useful single renal metric, **needs a medulla mask**. Do we (a) require medulla masks and accept that many users won't have them, (b) derive them from a T1 map when present, or (c) ship CMR as optional and accept that most reports won't have it? My inclination is (b) with (c) as fallback — but this determines whether the toolbox needs a T1-based segmentation step at all.
2. **Is CMR-as-mask-check the right framing, or is it too clever?** I've deliberately made it INFO/WARN-only and forbidden it from ever FAILing, because its denominator has an inter-session ICC of 0.08 at 3 T. Is a metric that can only ever flag *its own inputs* worth shipping, or does it just confuse a report? Is there a renal reader who would actually act on "CMR 1.18, masks may not be separated"?

Thresholds we cannot set alone

3. **Whose outlier rule, and what rejected-pair count is a FAIL?** Four published parameterisations exist (K7.2) and no paper says how many rejected pairs makes a scan unusable. Hartevelde's rule fires in ~2/3 of *healthy* datasets with a ceiling of 2 rejected pairs. Is " ≥ 3 rejected pairs $\rightarrow$ WARN, surviving pairs $< 20 \rightarrow$ FAIL" defensible, or is applying the 20-pair *acquisition* recommendation to *surviving* pairs a misuse of it? **Is there an unpublished in-house standard the renal community actually runs?** This is the single most valuable answer the renal specialist could give.
4. **Which kidney T1 for the M0 TR correction?** The consensus prescribes $SI_{PD}/(1 - \exp(-TR/T1, \text{tissue}))$ and then never supplies a value; cortex and medulla differ by ~400 ms and the published cortex range spans ~827–1406 ms across field strengths. Do we (a) apply a single default and document it, (b) refuse to correct and WARN with the magnitude under both compartments (my current proposal), or (c) require a T1 map whenever $TR < 5$ s?
5. **Should we ever compute RBF ourselves, or only consume it?** If we quantify, we inherit $\lambda = 0.9$ (an admitted brain-borrowed value), $\alpha = 0.95/0.85$ **which must be multiplied by 0.93 per BGS pulse**, and a BGS pulse count that is rarely recorded. The brain toolbox settled this as "QC layer, light derivation only" — is the same line right for kidney, given there is no renal pipeline to defer to?

Calibration and data

6. **Is there any renal ASL dataset with expert quality ratings, or a repeat-scan dataset?** Without labels there can be no renal QEI and no calibrated Stream-B threshold — that is the binding constraint on this entire design. Could PARENCHIMA, the ISMRM Renal MR Study Group, or the PARENCHIMA process paper's proposed "reference data ... benchmarks to verify implementations in new sites" supply one?
7. **What renal test data can we get, matching the three brain vendor cases?** The minimum useful spread is native + transplant, FAIR + PCASL, 1.5 T + 3 T. Without at least a FAIR and a PCASL case I cannot even test the branch gating in K5.2, let alone calibrate anything.
8. **Missing QUIPSS II — FAIL or WARN?** The consensus says bolus-width control "*must be used to quantify perfusion*" (R4.5, 100% agreement but **25% abstention**), yet concedes many published renal FAIR studies omit it and quantify by substituting T1 for T11, underestimating flow. A hard FAIL rejects much of the literature; a WARN may be too soft for a rule stated with "must". Which?

Metrics I am not sure are the right primitives

9. **Is there a renal spatial-heterogeneity metric worth having at all?** sCoV has zero renal precedent, and whole-kidney CoV is dominated by normal anatomy. Would a **cortex-restricted CoV** be meaningful, or is the SAMURAI-style **cortico-medullary CNR** the better renal primitive? Does the renal community have an accepted heterogeneity measure I haven't found?
10. **Which Nery Table-4 items are realistically present in real renal data?** Table 4 lists ~25 items and roughly 8 have no BIDS field at all (physiological triggering, fat suppression, pixel bandwidth, slice gap). Checking for items nobody records produces a wall of UNKNOWNS. **Which subset is worth enforcing?** And should laterality and native-vs-transplant be required fields in our own non-standard extension?

Two things I could not read or resolve

11. **Taso et al. 2023, Magn Reson Med 89(5):1754–1776 — the ISMRM Perfusion Study Group body-ASL "Grey Paper" — is paywalled and I have not read it.** Its abstract says the organ-by-organ review **starts with kidneys** and that "*summaries and recommendations of acquisition parameters (when appropriate) are provided for each organ*", and its senior author is the same as Nery 2020's. **Could Maria supply a copy, and does it supersede or refine any Nery 2020 parameter?** This is the largest known gap in the evidence behind Module K8.
12. **Is the transplant-vs-native perfusion offset established at all?** The one study reporting both (n=4 transplants vs n=7 natives, 17-year age gap) found transplants *higher*, contrary to the usual framing; and stable-allograft cortical RBF spans ~196–278 across five studies at nominally equivalent function. My current proposal is a transplant **geometry** profile with **no perfusion band** — is that too conservative, or exactly right?

Every threshold in this document carries a provenance tier. Where the honest answer is "nobody has published this", the document says so rather than inventing a band — that is the same policy as `THRESHOLD_PROVENANCE.md`, applied to a domain where it bites much harder.